Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University of Valladolid
Short name / acronym used in projects	UVa
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	Main institutional website and any relevant team / laboratory / company webpages
Country	Spain
Organisation type	☒ Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Street, city, postcode, country Campus Miguel Delibes. Paseo de Belén, 17. 47011 Valladolid, Spain
Department(s), institute(s), research group(s) or unit(s) that would be involved	Please name the unit(s) most relevant to the potential MSCA-SE action Ocular Surface Research Group, Institute of Applied Ophthalmobiology (IOBA), University of Valladolid.
Additional institutional resources relevant to collaboration	e.g. core facilities, clinical units, technology platforms, animal facilities, data infrastructures Facilities and equipment's of the IOBA-UVa (https://www.ioba.es/investigacion/laboratorios-y-unidades-de-investigacion-clinica/) *and* centralized services of UVa (https://aplicaciones.educa.jcyl.es/infrared/welcome.action).

Main contact person for this profile	Name, title, position, department / unit Margarita Calonge, Full Professor of Ophthalmology, Director of IOBA, UVa.
Main contact email	mcalonge@uva.es
Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☒ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☒ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☒ Advanced therapies / regenerative medicine	☒ Gene therapy / cell therapy / tissue engineering
☒ Inflammation / biomarkers / neuroprotection	☒ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☒ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☒ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	e.g. retinal organoids; ocular surface inflammation; zebrafish; AI diagnostics; drug delivery Ocular surface inflammation; cell therapy; regenerative medicine; ocular pain; mesenchymal stem cells; clinical trials; secretome; extracellular vesicles; nanomedicine
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus

Please describe the main focus of the team / unit that could participate. Keep it concise but specific (recommended: 150-250 words).

Our team focuses on the translational research and development of advanced therapies for severe inflammatory ocular surface diseases (OSD), particularly limbal stem cell deficiency (LSCD), severe dry eye disease (i.e. Sjögren syndrome, ocular graft-versus-host disease (oGVHD)), and also as novel therapy for corneal neuropathic pain. The group combines expertise in ophthalmology, stem cell biology, regenerative medicine, immunology and nanobiotechnology to develop innovative therapeutic strategies to treat OSD and reduce vision loss.

Our current research is mainly focused on mesenchymal stromal/stem cell (MSC)-based therapies, including the development of genetically modified and nanoengineered MSCs with enhanced migratory, immunomodulatory and regenerative properties. In collaboration with expert groups in nanomedicine, we are developing cell-based and cell-free therapeutic approaches using nano-engineered MSCs as targeted drug delivery systems for ocular surface inflammation.

Our work integrates *in vitro* inflammatory models, preclinical animal models and translational studies with strong focus on clinical application (having successfully translated some therapies to the clinical field). The team has extensive experience in advanced therapy medicinal products (ATMPs), ocular surface regeneration, subconjunctival cell delivery and translational ophthalmology, and actively collaborates with clinical, industrial and regulatory partners.

The group is particularly interested in interdisciplinary collaborations involving regenerative medicine, nanotechnology, gene and cell engineering, immunomodulation, epithelial-immune interactions and cell-free therapeutic strategies for inflammatory OSD and neuropathic corneal pain.

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Our team could bring strong added value to a translational eye-health and vision consortium through its multidisciplinary expertise in regenerative medicine, stem cell biology, nanotechnology and translational ophthalmology. We have extensive experience in the development of advanced therapy medicinal products (ATMPs) for ocular surface diseases, integrating *in vitro* models, clinically relevant preclinical models and translational approaches with direct clinical applicability.

A key contribution of our group is the development of innovative mesenchymal stromal/stem cell (MSC)-based therapies, including genetically modified and nanoengineered MSCs with enhanced immunomodulatory and regenerative properties. In collaboration with experts in nanomedicine, we have developed cell-based targeted drug-delivery strategies and are currently exploring cell-free therapeutic approaches using MSC-derived secretome. These technologies may be highly relevant not only for ophthalmology, but also for a broader range of inflammatory and degenerative diseases, opening opportunities for cross-disciplinary innovation.

The group also possesses expertise in subconjunctival administration as a minimally invasive delivery route, ocular surface inflammation, epithelial-immune interactions and the translational validation of regenerative therapies. Importantly, our experience extents the full translational pathway, from experimental development and preclinical validation to clinical application and collaboration with industrial partners. The group also have clinicians expert in the diseases.

We believe this combination of ophthalmology, regenerative medicine, nanotechnology and translational research could contribute to generating innovative and clinically relevant solutions within a highly interdisciplinary consortium.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- *Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.*

- Development of advanced therapy medicinal products (ATMPs) for ocular surface diseases.
- Mesenchymal stromal/stem cell (MSC) biology, expansion and characterisation from adipose tissue and bone marrow sources.
- Genetic modification and nanoengineering of MSCs to enhance migratory, immunomodulatory and regenerative properties.
- Development of MSC-derived secretome (extracellular vesicles approaches for inflammatory disease modulation.
- *In vitro* models of ocular surface inflammation and epithelial-immune interactions.
- Multiparametric flow cytometry and spectral cytometry for immunophenotyping and receptor analysis.
- Multiplex cytokine and inflammatory biomarker profiling (Luminex technology, ELISA).
- Clinically relevant rabbit models of limbal stem cell deficiency (LSCD) and ocular surface regeneration.
- Expertise in minimally invasive subconjunctival cell administration.
- Histology, immunofluorescence and tissue characterisation of the ocular surface.
- Translational ophthalmology, including preclinical validation and clinical implementation of cell-based therapies.
- Experience in investigator-driven clinical trials, compassionate-use therapies and interaction with Spanish Agency of Medicines and Medical Devices (AEMPS).
- Collaboration with GMP-compliant cell production facilities, CROs and industrial partners.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities

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-Two Cell culture laboratories: equipped with three laminar flow hoods, three CO_2 incubators, two thermostatic bath, two inverted phase contrast microscopes, and two centrifuges.

- Experimental ocular pathology laboratory: equipped with everything necessary for the processing and analysis of histological samples for light microscopy (regular and special stains, immunohistochemistry, etc.).
- Fluorescence microscope with image capture and treatment system (Leica DM6000B, Wtzlar, Germany).
- Fluorescence automated upright microscope DM6B THUNDER (Leica)
- Access to a Leica TC SP8 Lightning confocal microscope at the Advanced Optical Microscopy Service Instrumental Techniques Laboratory of the University of Valladolid.
- SpectraMAx M5 spectrophotometer (Molecular Devices).
- Real time PCR system PCR 7500 System (Applied Biosystems).
- Heidelberg Retina Tomograph III in combination with the Rostock Cornea Module: (IOBA-UVa)
- Swept-source optical coherence tomography technology (Plex Elite; Carl Zeiss Meditec): (IOBA-UVa)
- Ocular Surface and Immunology Clinical Unit: this is the clinical unit of IOBA where ocular surface disease patients, are diagnosed and treated.
- OculoFacial Pain Clinical Unit: this is the clinical unit at IOBA where patients with severe ocular chronic pain are diagnosed and treated

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

Inbound researchers joining our research group will receive multidisciplinary training in the field of ocular surface regenerative medicine and advanced cell therapy. The host institution offers a highly translational environment integrating basic research, GMP-oriented manufacturing approaches, preclinical development and clinical application. Researchers will have the opportunity to acquire and strengthen the following competencies:

Scientific and technical skills:

Training in mesenchymal stem cell and epithelial cell culture, ex vivo models, nanoengineering, microscopy, molecular and cellular biology techniques, immunohistochemistry, and experimental in vivo models of ocular surface diseases.

Clinical and translational skills:

The ophthalmologists of this group have wide experience in severe diseases affecting the ocular surface and have already participated in the translational aspects of the research, already treating patients with stem cell therapy.

Regulatory and quality-management skills:

Introduction to European regulatory frameworks governing Advanced Therapy Medicinal Products (ATMPs), ethical requirements, quality assurance and GMP-related standards relevant to cell-based therapies.

Data analysis and digital skills:

Image analysis, experimental design, data reproducibility and management of biomedical research datasets.

Transferable and managerial skills:

Development of scientific communication abilities through manuscript preparation, conference presentations and interdisciplinary teamwork.

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

Participation in the programme would provide significant opportunities to strengthen the scientific, translational and collaborative capacity of our research group beyond our current expertise in ocular surface cell therapy.

We expect to benefit from the acquisition of advanced experimental approaches, bioengineering strategies, inflammation *in vitro* and *in vivo* models, organ-on-chip systems, organoids development and new technologies that could be adapted to regenerative ophthalmology and cell therapy applications.

Additionally, we expect to get a cross-disciplinary knowledge exchange by the interaction with experts from diverse biomedical and technological fields, including immunology, tissue engineering, bioinformatics, nanomedicine, advanced imaging, pharmacology and translational medicine.

We also expect to strengthen our translational research pipelines and gain further expertise in clinical trial design, patient-oriented validation strategies, and the clinical implementation of regenerative and advanced therapeutic approaches.

In addition, we expect to have more opportunities to incorporate computational tools, bioinformatics, machine learning and advanced data analysis approaches into experimental and clinical research workflows. And finally, it will help to generate high-impact joint publications, collaborative grant applications and shared training activities that will contribute to the long-term internationalisation and competitiveness of the research group.
Overall, participation would significantly reinforce the group's capacity to develop innovative regenerative medicine approaches while broadening its translational, technological and interdisciplinary reach beyond ophthalmology alone.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Margarita Calonge	Full Professor in Ophthalmology	Department of Surgery, Ophthalmology, ENT, and Physiotherapy, Medical School, UVa IOBA, UVa	Researcher	Experienced	mcalonge@uva.es
Ana de la Mata	Assistant Professor in Immunology	Department of Pediatrics, Immunology, Obstetrics and Gynecology, Nutrition and Food Science, Psychiatry and History of Science, Faculty of Medicine, UVa IOBA, UVa	Researcher	Experienced	ana.mata@uva.es
Teresa Nieto-Miguel	Associate Professor in Cell Biology	Department of Cell Biology, Genetics, Histology and Pharmacology, Faculty of Medicine, Uva IOBA, UVa	Researcher	Experienced	teresa.nieto@uva.es

Main disciplines represented in your team	e.g. ophthalmology, optometry, cell biology, immunology, genetics, biomaterials, data science, engineering, AI, regulatory affairs, marketing and commercialisation. Ophthalmology, cell biology, immunology, regenerative medicine, cell therapy, tissue engineering, stem cells.
Experience in EU / international / multi-partner collaborative projects	Please indicate the most relevant experience in 1 or 2 lines (no specific details here)
Experience with clinical, translational or patient-facing research (if relevant)	☒ Yes ☐ No ☐ Not applicable | Short note: Extensive experience in translational ophthalmology, including the development and clinical application of MSC-based therapies for ocular surface diseases. The IOBA-UVa Clinical Cell Therapy Unit has performed 97 stem cell transplants in patients with limbal stem cell deficiency, including clinical trials and compassionate-use treatments authorised by the Spanish Agency of Medicines and Medical Devices (AEMPS). The group published the first clinical trial showing efficacy of MSC for the human eye. The group also has expertise in minimally invasive

	subconjunctival cell administration and collaborates closely with industrial, regulatory and clinical partners.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

- Calonge M, Herreras JM, Pérez I, Fernández I, Nieto-Miguel T, López-Paniagua M, Galindo S, Alberca M, García-Sancho J, Sánchez A. *A proof-of-concept clinical trial using mesenchymal stem cells for the treatment of corneal epithelial stem cell deficiency*. Translational Research. 2019. 206: 18-40. https://doi.org/10.1016/j.trsl.2018.11.003

- Pérez I, Galindo S, López-Miguel A, Nieto-Miguel T, de la Mata A, López-Paniagua M, Alberca M, Herreras JM, Calonge M. *In vivo confocal microscopy in limbal stem cell deficiency after mesenchymal stem cell transplantation: A sub-analysis from a phase I-II clinical trial.* Ophthalmology and Therapy. 2023;12(6):3251-3262. https://doi.org/10.1007/s40123-023-00809-7

- Galindo S, López-Paniagua M, de la Mata A, Herreras JM, García-Vázquez B, Marceñido B, Rey E, Higuera-Barón C, Calonge M, Nieto-Miguel T. *Bone marrow- versus adipose tissue-derived mesenchymal stem cells for corneal failure in an experimental model of limbal stem cell deficiency.* Experimental Eye Research 2026;262:110737. https://doi.org/10.1016/j.exer.2025.110737

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). ***Indicate if international.***

- Allogeneic mesenchymal stem cells (MSCs) by ocular injection as therapy for graft-versus-host disease refractory to conventional treatments
Funding agency: Instituto de Salud Carlos III (ISCIII) – Independent Clinical Research Projects, Strategic Action in Health 2023
Reference number: ICI23/00027
Dates: 01/01/2024 – 31/12/2027
Role: Partner institution / Local Principal Investigator (Margarita Calonge, IOBA-UVa)
Coordinator: Instituto de Investigación Biomédica de Salamanca (IBSAL). PI: Fermín Sánchez Guijo
Participants: IBSAL, University Hospital of Salamanca, IOBA-University of Valladolid, Gregorio Marañón University Hospital
International: No

- Genetically modified mesenchymal stem cells to improve their potential as ocular surface cell therapy (GenMSCeye)
Funding agency: Spanish Ministry of Science and Innovation – Knowledge Generation Projects (2021 Call)
Reference number: PID2021-125142OA-I00
Dates: 01/09/2022 – 31/08/2026
Role: Beneficiary / Principal Investigators (Ana de la Mata and Alberto López-Miguel)
Funding: €157,300
International: No

- Nano-engineered mesenchymal stem cells to improve the efficacy of stem cell transplants to the ocular surface
Funding agency: Spanish Ministry of Science and Innovation / European Regional Development Fund. Knowledge Generation Projects Oriented to Social Challenges (2019 Call)
Reference number: PID2019-105525RB-100
Dates: 01/06/2020 - 30/11/2025
Role: Beneficiary / Principal Investigators (Margarita Calonge and Teresa Nieto-Miguel)
Participants: Institute of Applied Ophthalmobiology (IOBA), University of Valladolid; CIBER-BBN
International: No

- Aniridia: Networking to address an unmet medical, scientific, and societal challenge
Programme: European Cooperation in Science and Technology (COST Actions), Horizon 2020
Reference number: CA18116
Dates: 11/04/2019 – 10/04/2023
Role: Participant / Collaborating research group (Margarita Calonge, IOBA-UVa)
Coordinator: Neil Lagali (Sweden)

Participants: 27 European and associated countries
International: Yes
European networking action focused on rare ocular diseases, particularly aniridia and limbal stem cell deficiency, promoting collaboration in regenerative medicine, stem cell therapies, tissue engineering and translational ophthalmology.

- ERN-EYE: European Reference Network dedicated to Rare Eye Diseases
Programme: European Reference Network (ERN-EYE), European Commission
Dates: Ongoing (since 2022 participation as affiliated centre)
Role: Affiliated collaborating centre / Clinical and research partner
Coordinator: Prof. Hélène Dollfus (France)
Participants: 52 healthcare providers across 18 EU countries, plus affiliated partners from additional European countries
International: Yes
Participation through the Hospital Clínico Universitario de Valladolid and IOBA-UVa as associated centre. The network promotes cross-border collaboration in diagnosis, clinical management, research and clinical trials for rare eye diseases, including ocular surface disorders and limbal stem cell deficiency.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Please indicate only what can be shared at this stage. - Clinical Cell Therapy Unit with experience in translational implementation of ATMPs for ocular surface reconstruction. Close collaboration with GMP-compliant production facilities, CROs, industrial partners and regulatory agencies. - Experience in investigator-driven clinical trials and compassionate-use cell-based therapies for ocular surface reconstruction approved by the Spanish Agency of Medicines and Medical Devices (AEMPS). - Expertise in minimally invasive subconjunctival cell administration protocols administration protocols. - Development of clinically relevant rabbit models of limbal stem cell deficiency (LSCD) and ocular surface inflammation, including minimally invasive subconjunctival cell administration protocols. - European patent application related to nanoengineered mesenchymal stromal/stem cells (MSCs) and their therapeutic use as targeted drug-delivery systems for inflammatory and degenerative diseases.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☐ Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile.

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☐ Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name:
Date completed:

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)